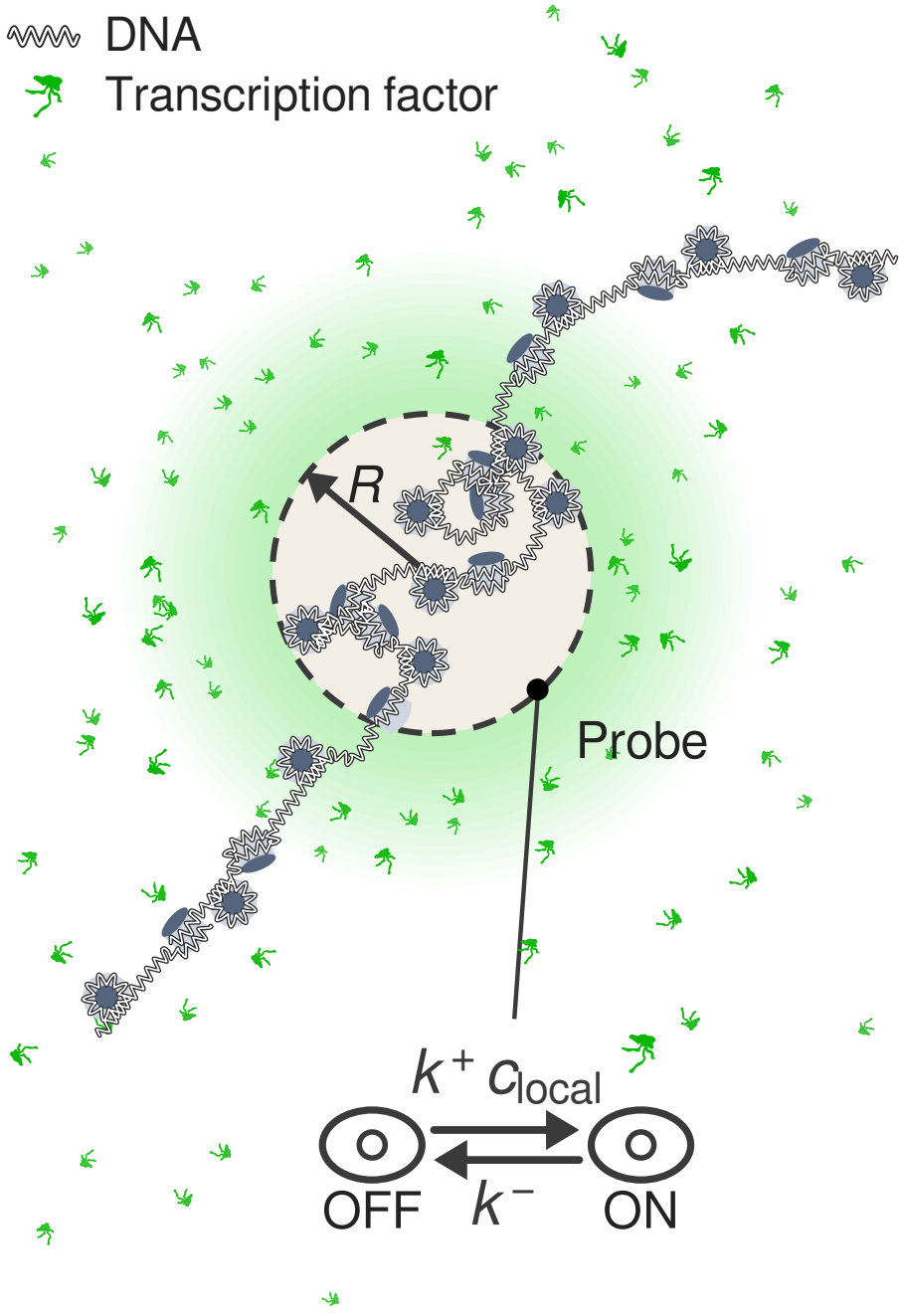
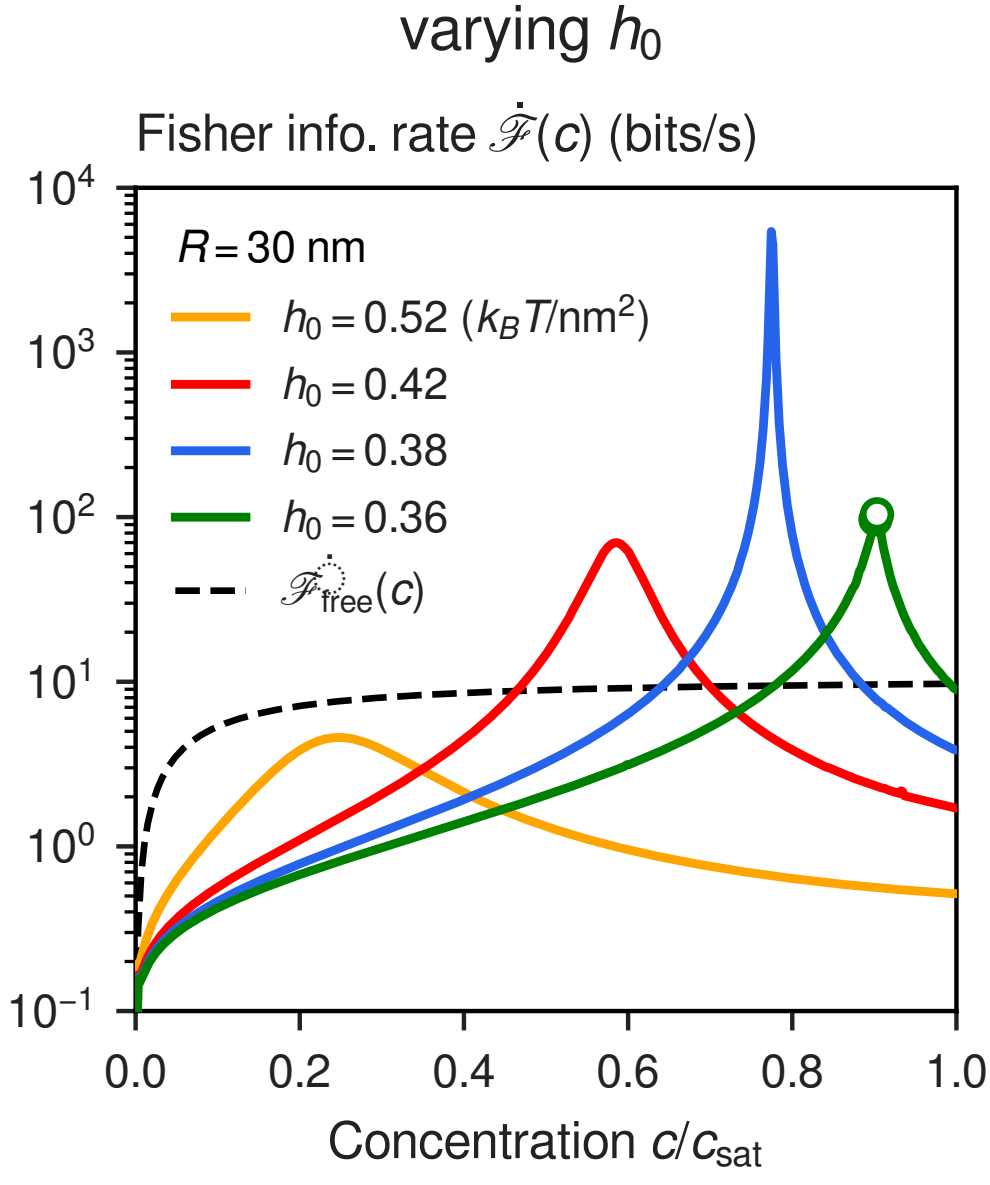

DNA
Transcription factor
R
Probe
$k^+ c_{\text{local}}$
k^-
OFF
ON


varying h_0
Fisher info. rate $\dot{\mathscr{F}}(c)$ (bits/s)
$R = 30$ nm
$h_0 = 0.52$ ($k_B T$/nm^2)
$h_0 = 0.42$
$h_0 = 0.38$
$h_0 = 0.36$
$\dot{\mathscr{F}}_{\text{free}}(c)$
10^4
10^3
10^2
10^1
10^0
10^{-1}
0.0
0.2
0.4
0.6
0.8
1.0
Concentration c/c_{sat}